

An Analysis of the Factors Influencing Residential Housing Prices Using Multiple Linear Regression

Predictive Analytics Short Case Data Storytelling

NITHYA BHAVANE D

2. Introduction

1.1 Background

One of the major financial decisions taken by a household is housing; the residential real estate market is notorious for prices being shaped by physical attributes like square footage, number of rooms, location, and amenities, which are not priced consistently across the market by different sellers, buyers, and agents. It's vital to understand which factors have an effect on price; this benefits multiple stakeholders like home buyers evaluating and negotiating price deals, sellers listing the price, realtors advising clients, and property developers deciding which amenities are worth building. That's the report's goal, i.e., to use these multiple features of housing and quantify their relationships rather than relying on intuition.

1.2 Dataset Description

The dataset used in "Housing Prices Dataset" is publicly available on Kaggle, containing 545 residential property records with 13 attributes, as it combines a continuous target variable price) with a mix of numeric and categorical predictors:

- Numeric: area (sq. ft.), bedrooms, bathrooms, stories, parking (number of spaces)
- Binary (yes/no): main road, guestroom, basement, hot water heating, air conditioning, preferred area.
- Categorical: furnishing status (furnished / semi-furnished / unfurnished)

The data contains no missing or duplicated values and required only categorical encoding before the model was run.

1.3 Objectives

- To explore the housing dataset and identify patterns and relationships between the attributes and price using Exploratory Data Analysis (EDA).

- To build a Multiple Linear Regression (MLR) model to quantify the effect of each attribute on housing price.
- To identify pricing-related problems or inefficiencies that the data reveals.
- To propose data-driven decision-making for various stakeholders in the market.

3. Data Insights

Exploratory Data Analysis was performed in Python through Google Colab using various libraries such as Pandas, seaborn, and matplotlib.

Table 3.1: Descriptive statistics

```

print("duplicate data:", df.duplicated().sum())
... duplicate data: 0

df.describe()
... Show hidden output

```

<i>Variable</i>	<i>Mean</i>	<i>Median</i>	<i>Std. Dev.</i>	<i>Min</i>	<i>25th Pct.</i>	<i>75th Pct.</i>	<i>Max</i>
<i>Price (INR)</i>	4,766,729	4,340,000	1,870,440	1,750,000	3,430,000	5,740,000	13,300,000
<i>Area (sq. ft.)</i>	5,150.54	4,600	2,170.14	1,650	3,600	6,360	16,200
<i>Bedrooms</i>	2.97	3	0.74	1	2	3	6
<i>Bathrooms</i>	1.29	1	0.50	1	1	2	4
<i>Stories</i>	1.81	2	0.87	1	1	2	4
<i>Parking spaces</i>	0.69	0	0.86	0	0	1	3

Distribution of House Price

Figure 3.1: Distribution of House Prices through graphical presentation

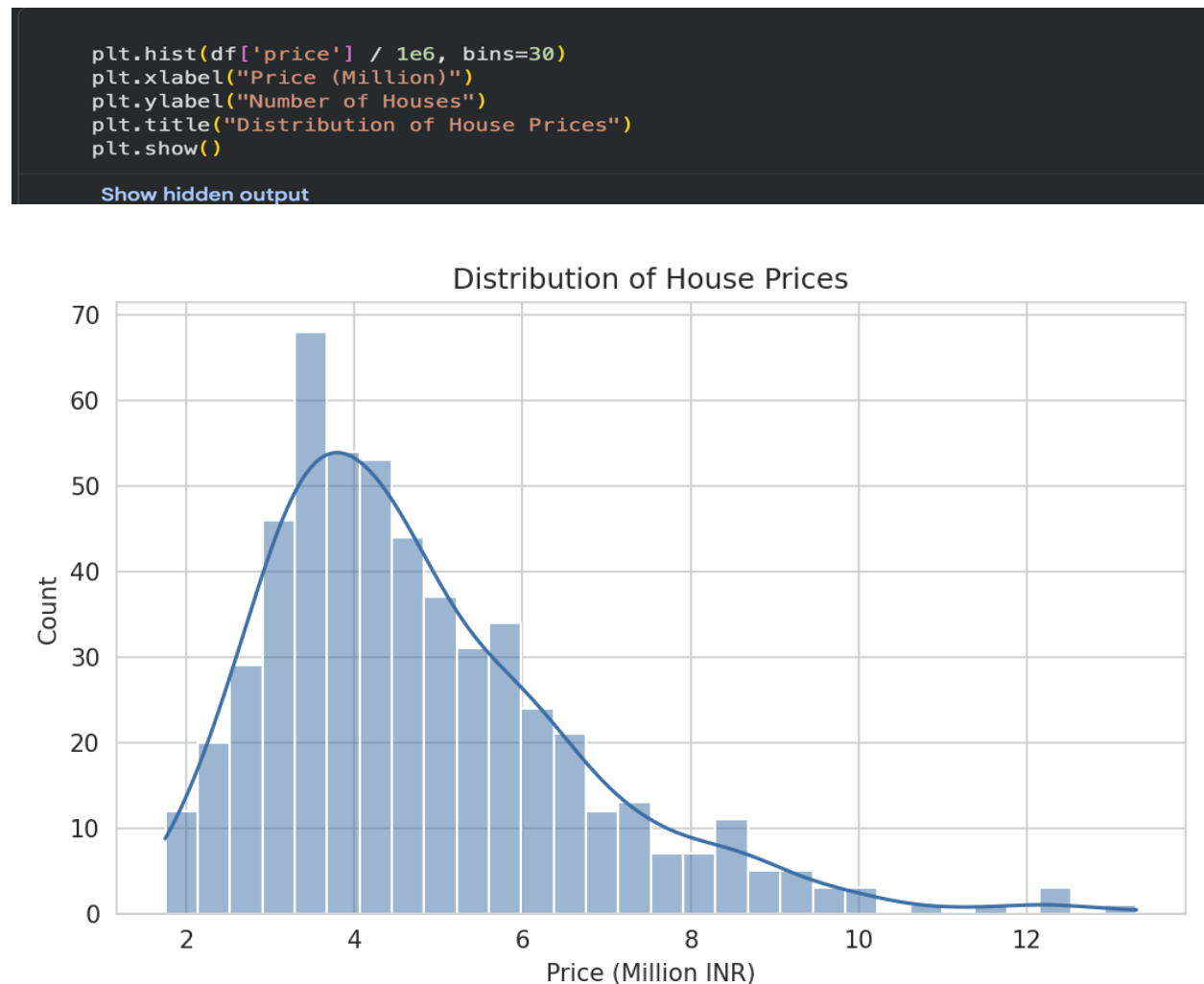
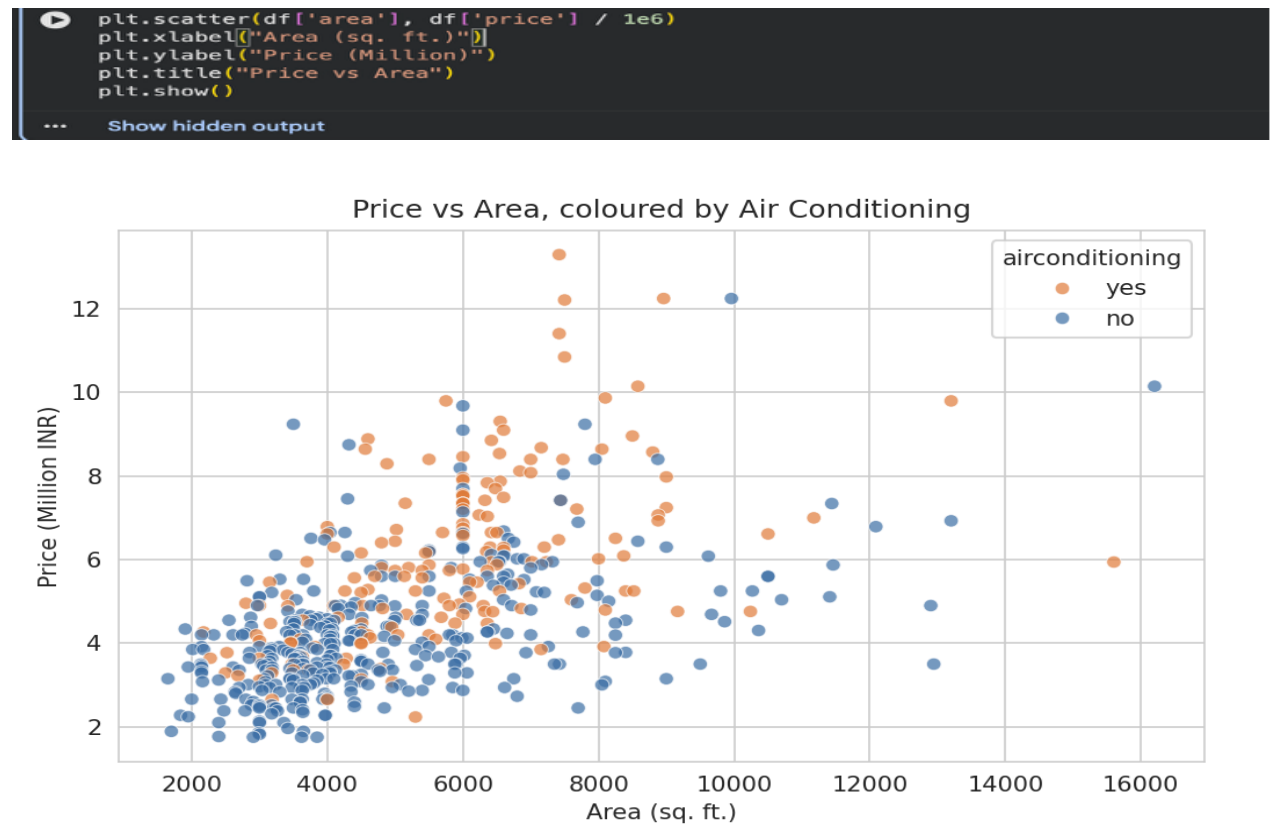


Figure 3.1 shows the distribution is rightly skewed which means that most houses are concentrated in the lower and middle price range, with a smaller number of houses in the higher range. The majority of the properties are priced around 3 to 6 million, with a few properties reaching above 10 million, which indicates the presence of some high-priced properties that may influence the average price. Table 3.1 summarizes the central tendency, where the average house price is about 4.77 million, while the median is 4.34 million, indicating that the prices are rightly skewed which can be seen in the scatter plot. The average area is about 5151 square feet, with properties ranging from 1650 to 16200 square feet. The average house has approximately 3 bedrooms, 1.3 bathrooms, 1.8 stories, and 0.69 parking spaces.

Price vs Area

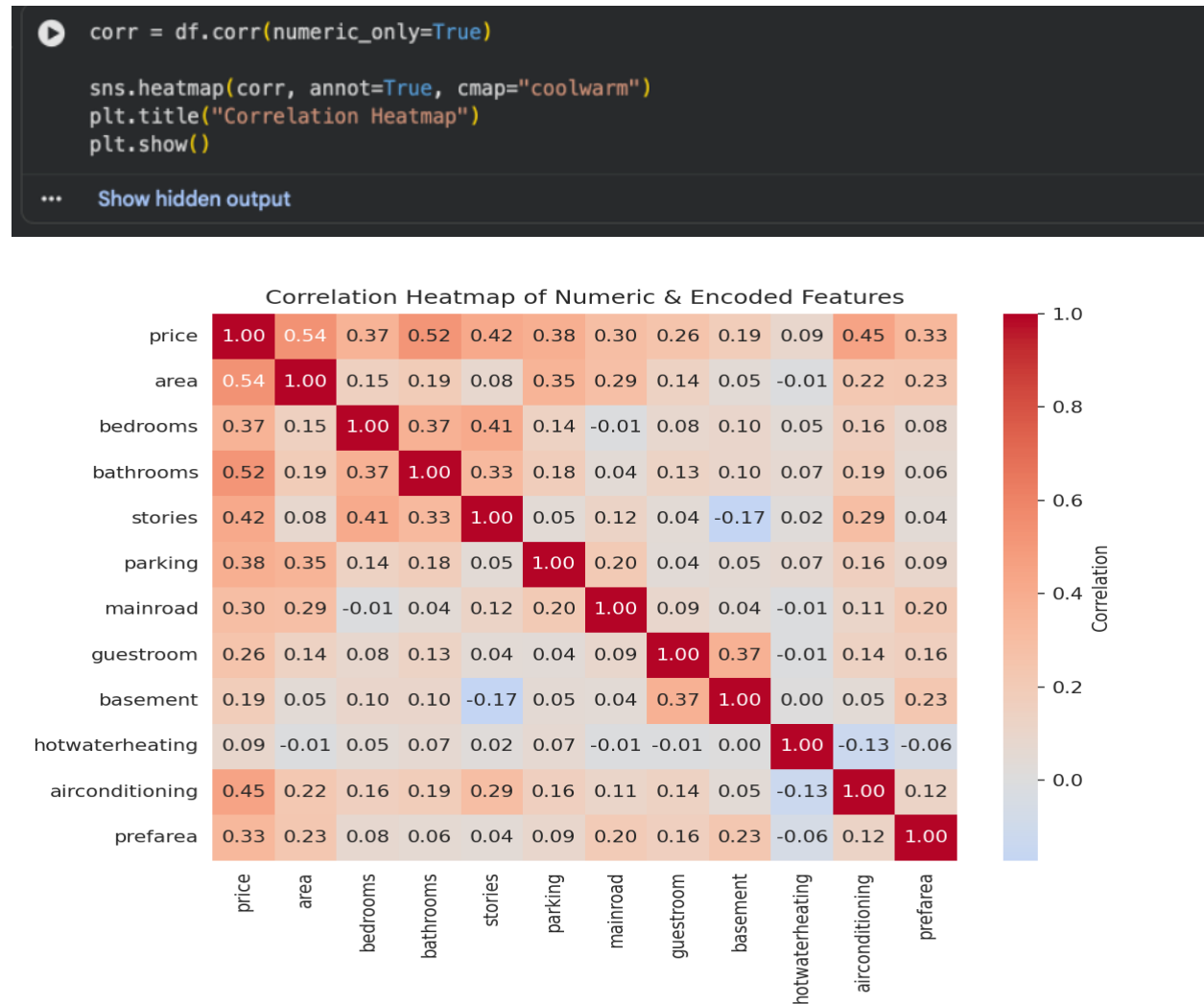
This scatter plot shows a positive relationship between area and price, where the larger houses generally tend to have higher prices, but it also shows that the houses with air conditioning tend to be associated with higher prices, although area alone does not completely determine price because there is considerable variation among properties of similar sizes.

Figure 2.2: Scatter plot representing Price vs. Area by Air conditioning



Correlation Structure

Figure 3.4: Encoded Features represented in a heatmap



The heatmap shows a correlation between price and binary variables, with area showing the strongest correlation, followed by bathrooms, air conditioning, stories, and bedrooms. It also shows relationships among predictors; for example, area, bedrooms, bathrooms, and stories are correlated with each other, indicating that multicollinearity is possible and might be a problem that can be considered in regression.

3.1 Problem Identification

- Amenity mispricing: Structural attributes such as air-conditioning, Heat water, and preferred location add premiums that are easy for a newbie, inexperienced seller to underprice or for a buyer to overlook when comparing listing prices on area or room counts.
- Bedroom count is an unreliable signal: Though no of bedrooms always makes it to the headline of listings, it's the smallest and statistically insignificant feature that has an effect on price. Hence, buyers who prioritize bedroom count over amenities might be overpaying.
- Unexplained price variance: the model explains about 68% of the price variation; the rest could be due to factors like neighborhood, property age and condition, and negotiation dynamics which aren't captured in the dataset.
- Multicollinearity risk: Bedrooms, bathrooms, area, and stories move together, which inflates the standard errors on individual coefficients and can make the model feature-level conclusions unstable if used without diagnosis.

3.2 Proposed solution

Multiple Linear Regression

To quantify the effects of the various attributes on price, a Multiple Linear Regression model was fitted with price as the dependent variable and all twelve remaining attributes as independent variables after encoding and one-hot encoding the furnishing status. Because multicollinearity was identified as a concrete problem in Section 3, it was corrected before finalizing the model, using a two-step remediation process rather than treating it as a mere caveat.

The first step was to target the variable; among the five high VIF variables, bedroom had both the highest VIF of 17.26 and a statistically significant coefficient in the initial OLS model, which indicates that it contributes to collinearity without contributing explanatory power; hence, it was dropped. Recomputing VIF on the remaining 12 predictors reduced every remaining value, bringing bathroom down from 10.12 to 8.49, area from 8.03 to 7.59, and stories from 7.99 to 6.34

```

201 X = df.drop(["price", "original_furnishingstatus"], axis=1)
    y = df["price"]

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )

    model = LinearRegression()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

211 print("R²:", r2_score(y_test, y_pred))
    print("MAE:", mean_absolute_error(y_test, y_pred))
    print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))

    R²: 0.6529242642153185
    MAE: 970043.4039201637
    RMSE: 1324506.9600914384

```

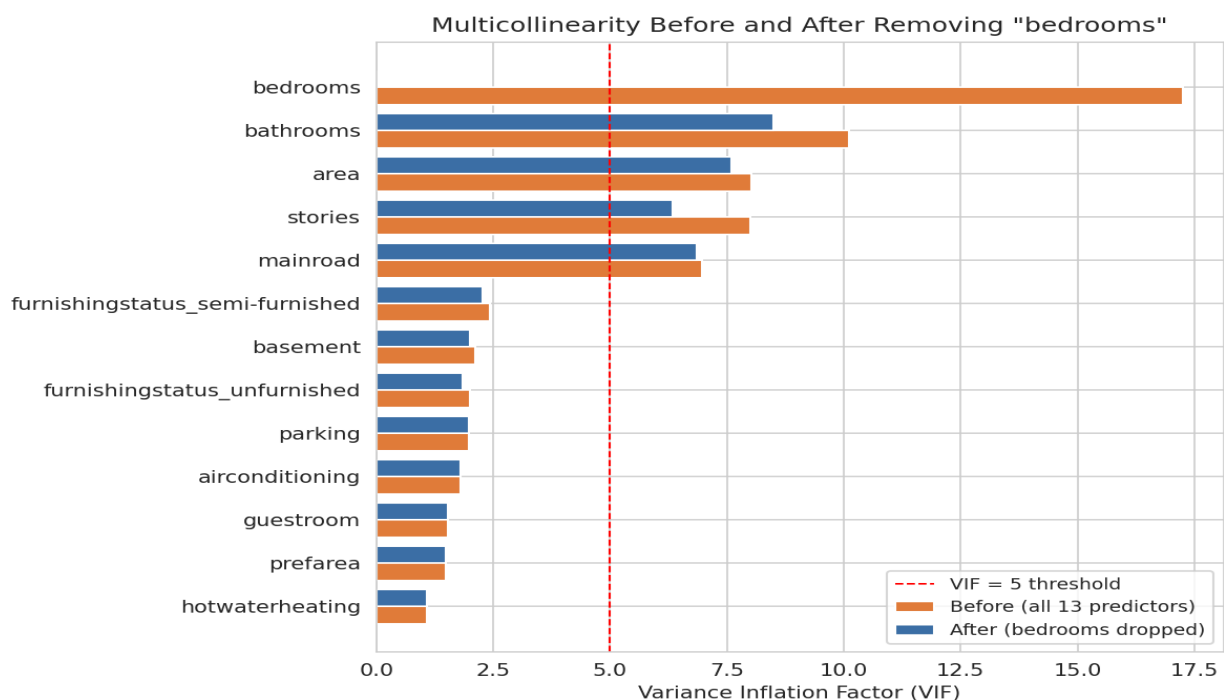


Figure 3.5: Multicollinearity before and after removing bedrooms

In the second step, regularization as the robustness was checked to further stabilize the remaining moderate collinearity among area, bathroom, stories, and main Rd. Where a Ridge regression model was fitted on standardized predictors, with the penalty strength selected. Ridge regression shrinks correlated coefficients towards each other instead of letting them grow to an extreme, which is the standard statistical process for dealing with multicollinearity when no variables should be dropped.

Table 3.2: Regression Model original vs correlated model

```

22] X_train_ols = sm.add_constant(X_train)
✓ Os  ols = sm.OLS(y_train, X_train_ols).fit()
      print(ols.summary())
>

```

Show hidden output

The table below compares the original model against the two corrected versions:

Model	Predictors	Train R^2	Test R^2	Test MAE (INR)
Original MLR (uncorrected)	12	0.686	0.653	970,043
MLR bedrooms removed	11	0.685	0.650	977,502
Ridge Regression (CV-tuned)	11 (standardised)	0.683	0.643	975,506

The table compares 3 models: the original MLR has a test R^2 of 0.653, while removing bedrooms gives a similar test R^2 of 0.650. Ridge regression gives 0.643, which indicates that removing bedrooms results in almost no loss of predictive performance while reducing the multicollinearity problem. Removing bedrooms due to collinearity virtually caused nothing in explanatory power, which confirms that bedrooms were redundant once area, bathroom, and storage were already included first. The Ridge model confirms that the corrected coefficient estimates are stable rather than problematic.

```

30] print("=== VIF BEFORE FIX ===")
6s print(compute_vif(X_train))

X_train_fixed = X_train.drop(columns=['bedrooms'])
X_test_fixed = X_test.drop(columns=['bedrooms'])

print("\n=== VIF AFTER FIX ===")
print(compute_vif(X_train_fixed))

X_train_sm_fixed = sm.add_constant(X_train_fixed)
model_final = sm.OLS(y_train, X_train_sm_fixed).fit()
print(model_final.summary())

lr_final = LinearRegression()
lr_final.fit(X_train_fixed, y_train)
y_pred_final = lr_final.predict(X_test_fixed)
print("Corrected Test R2:", r2_score(y_test, y_pred_final))
print("Corrected Test MAE:", mean_absolute_error(y_test, y_pred_final))

from sklearn.linear_model import RidgeCV
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_fixed)
X_test_scaled = scaler.transform(X_test_fixed)

ridge = RidgeCV(alphas=np.logspace(-2, 4, 100), cv=5)
ridge.fit(X_train_scaled, y_train)
print("Best alpha:", ridge.alpha_)
y_pred_ridge = ridge.predict(X_test_scaled)
print("Ridge Test R2:", r2_score(y_test, y_pred_ridge))

```

Table 3.3: MLR Metrics

Metric	Value
Train R^2	0.685
Adjusted R^2	0.676
Test R^2	0.650
Test MAE (INR)	977,502
Test RMSE (INR)	1,329,747

The model has a train R2 of 0.685, meaning it explains above 68.5% of the variation in the training data, while the adjusted R2 is 0.676, which accounts for the number of predictors. The tests are two of 0.650 means that the model explains about 65% of the variation in price, where the test MAE is approximately 977502, which means that the model corrections or, on average, about 9.78 lakh away from the actual prices. The final model explains roughly 2/3 of the price variation on the unseen data, which matches the original model performance while solving the collinearity problem, which indicates that this insight can be relied upon.

```
residuals = y_test - y_pred

plt.scatter(y_pred / 1e6, residuals / 1e6)

plt.axhline(0, color='red', linestyle='--')

plt.xlabel("Predicted Price (Million INR)")
plt.ylabel("Residual (Million INR)")
plt.title("Residuals vs Predicted Price")

plt.show()
```

Show hidden output

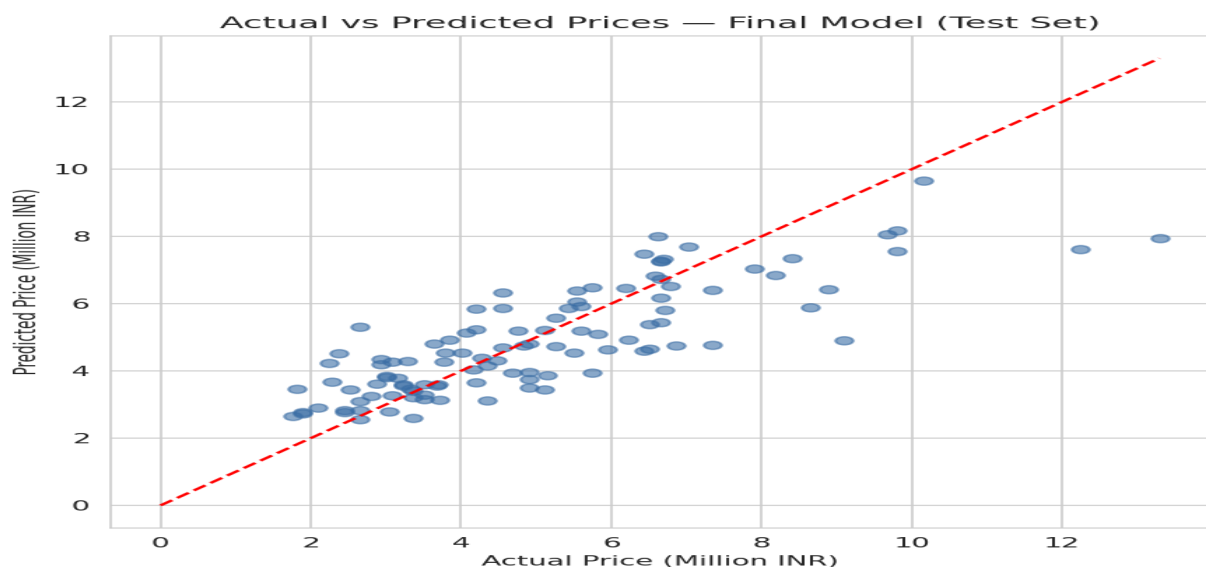


Figure 3.6: Actual vs. predicted prices on the test set from the final (corrected) model. Points close to the red diagonal indicate accurate predictions.

The graph compares the actual house prices with the prices predicted by the final MLR model, where the points are closer to the diagonal line, which represents more accurate predictions. Most points are reasonably close to the line, showing that the model captures a substantial part of the

variation in housing prices, although some predictions have visually noticeable errors, particularly for higher-priced houses.

4.3 Key Coefficients (per unit, holding other variables constant)

```
from sklearn.linear_model import RidgeCV
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_fixed)
X_test_scaled = scaler.transform(X_test_fixed)

ridge = RidgeCV(alphas=np.logspace(-2, 4, 100), cv=5)
ridge.fit(X_train_scaled, y_train)
print("Best alpha:", ridge.alpha_)
y_pred_ridge = ridge.predict(X_test_scaled)
print("Ridge Test R2:", r2_score(y_test, y_pred_ridge))

vif_before = compute_vif(X_train)
vif_after = compute_vif(X_train_fixed)
merged = vif_before.rename(columns={'VIF': 'VIF_before'}).merge(
    vif_after.rename(columns={'VIF': 'VIF_after'}), on='feature', how='left')
merged = merged.sort_values('VIF_before', ascending=True)
```

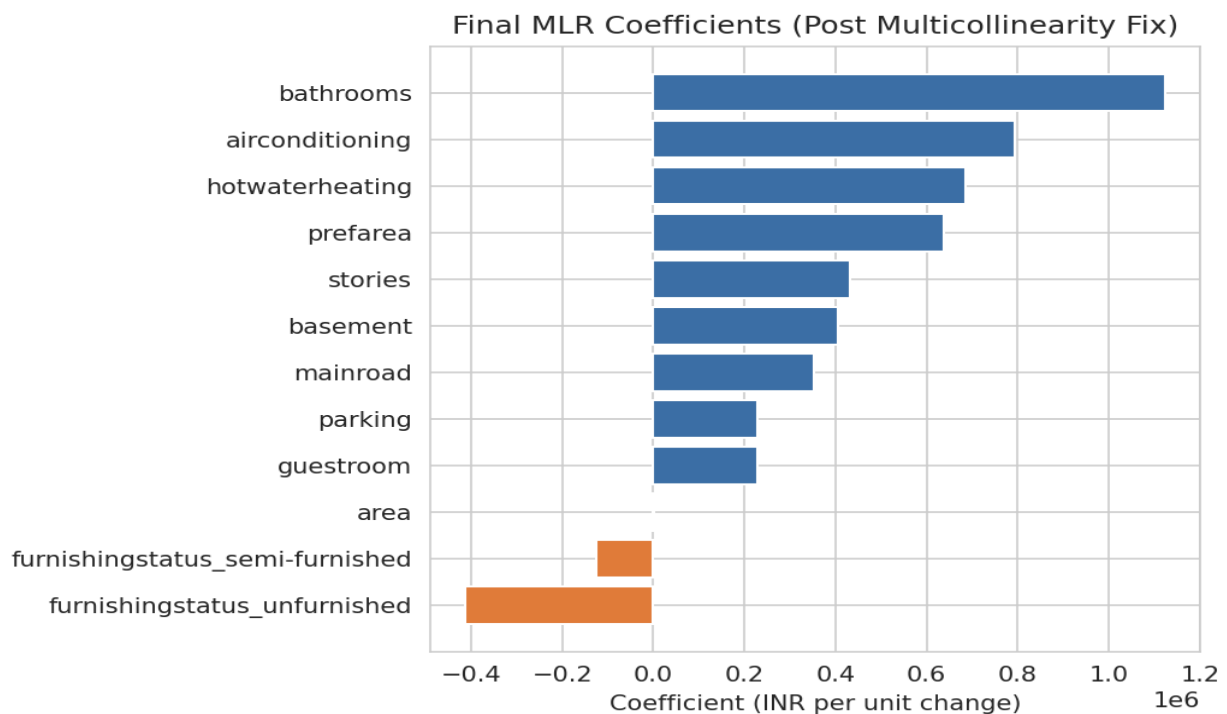


Figure 3.7: Coefficient plot from the final, multicollinearity-corrected MLR model (blue = increases price, orange = decreases price relative to the reference category).

The coefficient plot shows how each predictor affects the housing price while keeping the other variables constant. The blue bars represent positive effects, while the orange bars represent negative effects. Here, bathrooms have the largest positive coefficient, followed by air conditioning, hot water heating, preferred areas, and storage, while unfurnished status has a negative coefficient; unfurnished houses tend to have lower prices than furnished houses.

Table 3.4: Independent Features and Coefficients

```
 coefficients = pd.DataFrame({
    "Variable": X.columns,
    "Coefficient": model.coef_
})

print(coefficients.sort_values("Coefficient", ascending=False))
```

Variable	Effect on Price (INR)	Statistically Significant?
Bathrooms (+1)	+ 1,123,906	Yes (p < 0.001)
Air-conditioning (yes)	+ 792,643	Yes (p < 0.001)
Hot-water heating (yes)	+ 685,631	Yes (p = 0.002)
Preferred area (yes)	+ 636,490	Yes (p < 0.001)
Stories (+1)	+ 431,258	Yes (p < 0.001)
Basement (yes)	+ 406,465	Yes (p = 0.001)
Main road access (yes)	+ 351,400	Yes (p = 0.018)
Parking (+1 space)	+ 228,672	Yes (p < 0.001)
Guestroom (yes)	+ 227,452	No (p = 0.106)
Area (+1 sq. ft.)	+ 238	Yes (p < 0.001)
Unfurnished (vs. furnished)	- 413,335	Yes (p = 0.002)

The table identifies which variables significantly influence housing prices; bathrooms have the largest positive effect, with one additional bathroom associated with approximately 11.24 lakh higher price. Meanwhile, air conditioning, hot water heating, preferred area, stories, basement, main road access, parking, and area also have a positive significant effect, but guest rooms are not statistically significant because its P value is 0.106. Unfurnished houses are associated with approximately 4.13 lakh lower prices compared with furnished houses.

These results translate into recommendations that are backed by data:

- For sellers: Prioritize disclosing and highlighting bathrooms, air conditioning, hot water heater, and preferred area in listings, as these carry the largest statistical significance in price premiums and are larger than an additional bedroom.
- For Buyers: Do not over-index on bedroom count when comparing similarly priced homes; area, bathroom counts, and amenities are far stronger value drivers, so a small bedroom home with better amenities may be a better deal per value you pay. \
- For agents: from the analysis, it is found that unfurnished homes are sold for 4.1 lakh less than furnished houses; hence, the furnishing status matters; hence, light furnishing before listing could be a low-cost way to lift price.

3.3 Conclusion

This case study applied multiple linear regression to the Kaggle dataset about housing prices to identify which property attributes most strongly influence prices. The model explains approximately 65 to 69% of price variation and confirms that bathrooms, air conditioning, hot water heating, preferred area location, and number of stories are the strongest statistically significant drivers of price, while bedrooms had high collinearity and had to be removed.

These prices carry practical implications where price and purchase decision in the market should weight amenities usually space more heavily than headline bedroom counts, an automated valuation model built on this model should address the multicollinearity among size related variables before being used for real pricing decisions as the unexplained 32 to 35% of the variation also underscores the location specific and qualitative factors which were not present in the data and remain important should be incorporated in later in the future during data collection.

Overall, the analysis demonstrates how a relatively simple regression model can convert a raw housing data set into meaningful insights and then actionable guidance for buyers, sellers, and agents who are making decisions in the market.

Kaggle account: <https://www.kaggle.com/nithyabhavanerajan>